

newton's laws of motion - practice problems

name _____

1. An 8.0 kg crate of glassware is pushed to the right with a 30 N force. Friction pushes left on the crate with a 6.0 N force. What is the crate's acceleration?

2. A 4.0 kg cart starts from rest on a 2.0 m level track. A net force of 10 N pushes it to the right for 1.2 s. What is its acceleration, and how far does it move?

3. Two students pull on a 15 kg cart. One pulls left with 40 N and the other pulls right with 40 N. What is the cart's acceleration?

4. A student pushes a loaded wagon with a net force of 72 N, and it accelerates at 3.0 m/s^2 . What is the mass of the wagon and its load?

5. A 12 kg box of books is pushed across a carpeted floor with a 40 N horizontal force. What is the box's acceleration?

6. A 12 kg sled accelerates to the right at 2.0 m/s^2 . Friction pushes left on the sled with 5.0 N. What applied force to the right is needed?

7. A 4.5 m canoe and its paddler have a total mass of 90 kg. While the paddler rests, the wind pushes the canoe left with 60 N and the current pushes it right with 42 N. What is the canoe's acceleration?

8. A 3.0 kg delivery drone has a 30 N force on it to the right, but it accelerates to the left at 4.0 m/s^2 . What leftward force must also be acting on it?

9. A 1500 kg car cruises down a straight highway at a constant 25 m/s. The engine pushes the car forward with 600 N. What is the net force on the car, and how large are the air and road resistance forces?

10. A student throws a 500 g ball, and the ball accelerates to the right at 30 m/s^2 while it's in the student's hand. What force does the student apply to the ball?

11. A mosquito hits the windshield of a truck on the highway. Which feels a larger force, the mosquito or the truck? Which has a larger acceleration?

12. Two ice skaters stand facing each other and push off. Skater A has a mass of 60 kg and skater B has a mass of 45 kg. During the push, each skater feels a 90 N force. Skater A moves left and skater B moves right. What is each skater's acceleration?

13. A 12 kg backpack falls off a ledge. Ignoring air friction, it accelerates downward at 9.8 m/s^2 , like everything in free fall on Earth. What net force causes that acceleration?

14. A 900 kg car has a 3.2 kN engine force forward and 0.80 kN of air and road resistance backward. What is the car's acceleration?

15. A 1200 kg car goes from rest to 100 km/h in 9.3 s. What average net force acts on the car?

16. An empty 10 kg shopping cart accelerates at 2.0 m/s^2 when you push it. You load it with 15 kg of groceries and push with the same force. What is the new acceleration?